MD Jakir Hossain

Website | Google Scholar | GitHub | LinkedIn

Phone: (+880) 1742750366 | Email: mdjakirhossen13@gmail.com

RESEARCH PROFILE

Machine learning researcher with a publication record of 17+ peer-reviewed works and hands-on experience building and rigorously evaluating deep learning systems for medical and biomedical applications. Core strength lies in transfer-learning-based model development (CNNs, EfficientNet, ResNet, Vision Transformers), explainable AI (Grad-CAM) for auditing model decision-making, and building reproducible, version-controlled research pipelines for large-scale data preprocessing, training, and cross-dataset evaluation. Recent work extends this foundation into large language models for healthcare – fine-tuning Qwen2.5-1.5B with QLoRA on the PubMedQA biomedical question-answering dataset – reflecting a growing interest in applying foundation models to structured clinical and health-related data. Motivated to build on this methodological foundation in machine learning, reproducible research practice, and medical AI to pursue graduate research in health data science, including the use of large-scale health data and modern AI methods (LLMs, foundation models) for clinical prediction.

RESEARCH INTERESTS

Machine Learning and Deep Learning for Healthcare, Explainable AI (XAI), Transfer Learning, Large Language Models and Foundation Models for Clinical Applications, Medical Image Analysis, Reproducible and Robust Research Pipelines, Applied Data Science, Health Data Science.

EDUCATION

East West University (EWU), Dhaka, Bangladesh — *Jan 2024 – Feb 2025*
MS in Computer Science and Engineering — GPA: **3.75 / 4.0**
Major: Data Science

East West University (EWU), Dhaka, Bangladesh — *Jan 2019 – May 2023*
B.Sc. in Computer Science and Engineering — GPA: **3.53 / 4.0**
Major: Intelligent Systems and Data Science

RESEARCH EXPERIENCE

Research Assistant — February 2024 – February 2025
Supervisor: Prof. Dr. Ahmed Wasif Reza — East West University, Dhaka, Bangladesh

- Designed and benchmarked deep learning architectures (CNNs, Vision Transformers, attention-based multimodal models) across medical, dermatological, and hematological imaging domains, building broad hands-on expertise in model design, training, and rigorous empirical evaluation.
- Applied Grad-CAM-based explainability techniques to audit decision-making in vision and language models, developing an interpretability-first evaluation methodology directly relevant to trustworthy, auditable AI in clinical settings.
- Engineered reproducible, GPU-accelerated pipelines for large-scale data preprocessing, training, and cross-dataset evaluation, emphasizing methodological robustness and version-controlled workflows – practices directly transferable to large-scale health data research.
- Collaborated within interdisciplinary research teams to design, execute, and publish 7+ peer-reviewed studies spanning medical image analysis and biomedical classification, contributing to study design, statistical evaluation, and scientific writing.

MASTER'S THESIS

Diabetic Retinopathy Detection using Attention-Enhanced Transfer Learning and Explainable AI Techniques
Supervisor: Dr. Mohammad Rifat Ahmmad Rashid, East West University

Developed a reproducible deep learning pipeline combining attention-enhanced transfer learning (ResNet, EfficientNet) with merged multi-source image datasets to improve cross-dataset generalization for a clinically relevant disease-classification task. Integrated Grad-CAM-based explainable AI to audit model decision-making and localize disease-relevant regions – establishing a systematic framework for model transparency and reliability that generalizes beyond imaging to any domain requiring interpretable, clinically trustworthy predictions.

SELECTED RESEARCH PROJECTS

Medical LLM Fine-Tuning using Qwen2.5-1.5B & QLoRA 2026
GitHub | *Hugging Face* *(Ongoing)*

- Fine-tuned the Qwen2.5-1.5B language model using QLoRA (quantized low-rank adaptation) on the PubMedQA biomedical question-answering dataset, applying parameter-efficient fine-tuning to a domain-specific clinical NLP task.
- Built an end-to-end, reproducible training pipeline using Hugging Face Transformers, PEFT, TRL, and BitsAndBytes, and published the resulting model on Hugging Face with version-controlled, reproducible research code on GitHub.

Multimodal Deep Learning Framework for Neurological Disease Classification (PDNet) 2026
github.com/jakirniloy/PDNet-Model-for-Parkinson-s-Disease *(Submitted, Applied AI Letters)*

- Designed a multimodal fusion architecture integrating structural and functional imaging inputs with convolutional feature extraction and transfer learning for disease classification.
- Built a reproducible, version-controlled pipeline for data preprocessing, model training, and evaluation.

Deepfake Video Detection Using Vision Transformers 2026
github.com/jakirniloy/Deepfake-Video-Detection-Using-Vision-Transformers *(Ongoing)*

- Designed ModifiedViT, a 24-block Vision Transformer with 1024-dimensional embeddings, applying a transfer-learning and robustness-evaluation methodology consistent with the medical-imaging work above – a generalizable, dataset-agnostic approach to evaluating deep learning systems.

AgriFreshNet: AI-Powered Fruit Detection & Freshness Classification Submitted, Journal of Food Process Engineering
github.com/jakirniloy/AgriFreshNet | *Dataset*

- Developed a real-time computer vision pipeline for fruit detection and freshness classification using EfficientNet-based CNN models in PyTorch, deployed via a Streamlit application – demonstrating end-to-end AI system deployment.

PUBLICATIONS

Selected: Medical & Biomedical AI

[M1] M. J. Hossain *et al.*, "Enhancing Skin Cancer Detection Through Automated Classification Using Convolutional Neural Networks," *ICO 2023, LNNS*, vol. 1167, Springer, Cham, 2024. DOI: 10.1007/978-3-031-73318-5_10 | Code

[M2] N. N. Juthi, T. S. Mahi, . . . , M. J. Hossain *et al.*, "Dermatological Disease Detection Using Image Processing and Deep Learning," *2024 IEEE COMPAS*, 2024, pp. 1–5. DOI: 10.1109/COMPAS60761.2024.10797116

[M3] R. Jahan, A. Sarker, . . . , M. J. Hossain *et al.*, "Developing a Model for Predicting Multiple Sclerosis Progression Using MRI Scans," *2024 IEEE COMPAS*, 2024, pp. 1–5. DOI: 10.1109/COMPAS60761.2024.10796262

[M4] M. Wahiduzzaman, M. J. Fokir, . . . , M. J. Hossain *et al.*, "Utilizing Deep Learning for Detecting Malaria in Blood Cell Images," *2024 IEEE COMPAS*, 2024, pp. 1–5. DOI: 10.1109/COMPAS60761.2024.10796987

Applied Machine Learning, Transfer Learning & NLP

[A1] M. J. Hossain *et al.*, "Enhancing Fake News Detection Through Machine Learning and Transfer Learning Methods," *ICDMIS 2024, LNNS*, vol. 1386, Springer, 2026. DOI: 10.1007/978-981-96-6053-7_28

[A2] M. J. Hossain *et al.*, "Deciphering Textual Emotions: Sentiment Analysis with Comprehensive Improvement Through Transfer Learning," *ICDMIS 2024, LNNS*, vol. 1387, Springer, 2026. DOI: 10.1007/978-981-96-6060-5_8

[A3] M. J. Hossain, S. S. Zaman, F. R. Akash, M. R. Sarker, and M. R. Huq, "Developing a Bangla Handwritten Text Recognition Framework using Deep Learning," *2023 26th ICCIT*, Cox's Bazar, Bangladesh, 2023, pp. 1–6. DOI: 10.1109/ICCIT60459.2023.10441107

[A4] M. J. Hossain *et al.*, "Detection of Deceptive Hotel Reviews Through the Application of Machine Learning Techniques," *CIIR 2023, LNNS*, vol. 1025, Springer, 2024. DOI: 10.1007/978-981-97-3594-5_38

[A5] M. J. Hossain, F. R. Akash *et al.*, "Deciphering Handwritten Text: A CNN Framework for Handwritten Character Recognition," *ICO 2023, LNNS*, vol. 729, Springer, 2023. DOI: 10.1007/978-3-031-36246-0_18

[A6] A. Hossain, **M. J. Hossain**, I. A. Arin *et al.*, "Dry Fish Image Dataset: Data-Driven Analysis and Deep Learning-Based Classification," *Data in Brief*, vol. 66, p. 112683, June 2026. DOI: 10.1016/j.dib.2026.112683

[A7] M. M. Hassan *et al.* (incl. M. J. Hossain), "Smart Spectacles for The Deaf with Voice to Text and Sign Language Integration," *2023 26th ICCIT*, 2023, pp. 1–6. DOI: 10.1109/ICCIT60459.2023.10441555

[A8] M. S. Islam *et al.* (incl. M. J. Hossain), "Deep Learning-Based Classification of Coconut Leaf Diseases Using Optimized CNN Models," *2025 QPAIN*, Rangpur, Bangladesh, 2025. DOI: 10.1109/QPAIN66474.2025.11172002

[A9] M. S. Islam *et al.* (incl. M. J. Hossain), "Detection of Sugarcane Leaf Diseases Using Custom CNN and ResNet50," *ICIDA 2024, LNNS*, vol. 1410, Springer, 2025. DOI: 10.1007/978-981-96-6303-3_20

[A10] M. S. Islam *et al.* (incl. M. J. Hossain), "Earthquake Magnitude Prediction: A Survey on ML Models, Datasets, Techniques, Challenges, and Future Directions," *ICIDA 2024, LNNS*, vol. 1410, Springer, 2025. DOI: 10.1007/978-981-96-6303-3_42

[A11] S. S. Zaman *et al.* (incl. M. J. Hossain), "Measuring Accuracy of Different ML Algorithms on Expectations and Experiences of University Students," *2025 QPAIN*, Rangpur, Bangladesh, 2025. DOI: 10.1109/QPAIN66474.2025.11171720

Additional publications available on Google Scholar.

TECHNICAL SKILLS

Machine Learning & Deep Learning	CNNs, Vision Transformers (ViT), attention mechanisms, transfer learning, multimodal fusion, Transformers, Autoencoders, Generative Models (VAE/GAN)
LLMs & NLP	Hugging Face Transformers, PEFT/QLoRA, TRL, BitsAndBytes, parameter-efficient fine-tuning, domain-specific (biomedical) QA
Explainable AI (XAI)	Grad-CAM, SHAP (familiar); interpretable and reproducible model evaluation pipelines
Medical & Biomedical AI	Medical image classification (retinal, dermatological, hematological), cross-dataset generalization, biomedical question answering
Data Science & Reproducible Research	Pandas, NumPy, SciPy, PySpark, GPU-accelerated computing (CUDA/PyTorch), version-controlled, cross-dataset benchmarking pipelines
AI / DL Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, JAX (familiar)
Real-World AI Systems	Streamlit deployment, end-to-end inference pipelines
Programming	Python (primary), C#, Java, C, C++, SQL
Research Tools	Git, GitHub, reproducibility and robustness benchmarking across datasets

PROFESSIONAL EXPERIENCE

Junior AI Engineer — August 2025 – Present
Algoramming, Dhaka, Bangladesh — www.algoramming.com

- Investigated cross-modal alignment and consistency in multimodal deep learning pipelines, contributing to robustness and reliability assessment of production-grade model outputs.
- Designed and optimized deep learning architectures (CNNs, transformers, attention-based mechanisms), building and deploying end-to-end ML/DL pipelines with emphasis on scalability, reproducibility, and GPU-accelerated training using Python and PyTorch.
- Collaborated with cross-functional, interdisciplinary teams to integrate AI models into production systems, ensuring reliability and computational efficiency.

TEACHING EXPERIENCE

Graduate Teaching Assistant — June 2023 – July 2025
Department of Computer Science and Engineering — East West University, Dhaka, Bangladesh

- Supervised and mentored undergraduate students in the statistical and mathematical foundations of machine learning, probability, and linear algebra, providing structured feedback and guidance.
- Coordinated course logistics and led student presentations and discussion sessions – developing organizational and presentation skills transferable to research supervision and collaborative lab work.

SCHOLARLY IMPACT

Google Scholar Metrics

- **Citations:** 82
- **h-index:** 6
- **i10-index:** 4
- **Publications:** 17+ peer-reviewed

AD Scientific Index ID: 5909138

Rankings by Total H-Index:

- **World:** #1,882,954
- **Asia:** #525,464
- **Bangladesh:** #9,339
- **East West University:** #133

REFERENCES

Prof. Dr. Ahmed Wasif Reza
Professor & Dean, Faculty of Sciences and Engineering
Department of Computer Science & Engineering
East West University, Dhaka, Bangladesh
Email: wasif@ewubd.edu
Phone: +8801314447616

Dr. Mohammad Rifat Ahmmad Rashid
Associate Professor
Department of Computer Science & Engineering
East West University, Dhaka, Bangladesh
Email: rifat.rashid@ewubd.edu
Phone: +8801780099173